Deep Multilingual Correlation for Improved Word Embeddings

Ang Lu[1], Weiran Wang[2], Mohit Bansal[2], Kevin Gimpel[2] and Karen Livescu[2]
[1]Tsinghua University [2]Toyota Technological Institute at Chicago

Overview

- word embeddings can be improved via multilingual correlation using CCA [1]
- we use **Deep CCA** [2], permitting the correlated information to lie in nonlinear subspaces
- resulting monolingual embeddings improve performance in word and bigram similarity tasks

Multilingual Correlation

Find pairs of vectors for both input views that maximize correlation of projections

First pair:

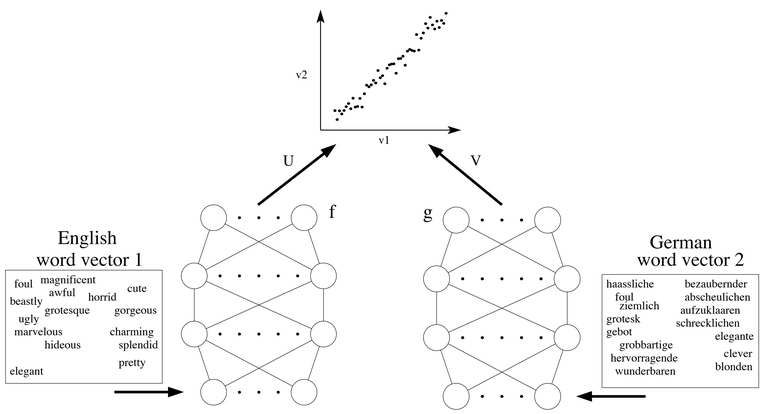
$$\max_{u \in \mathbb{R}^{D_x}, v \in \mathbb{R}^{D_y}} \frac{\mathbb{E}[(u^T x)(v^T y)]}{\sqrt{\mathbb{E}[(u^T x)^2]}\sqrt{\mathbb{E}[(v^T y)^2]}} = \frac{u^T \Sigma_{xy} v}{\sqrt{u^T \Sigma_{xx} u}\sqrt{v^T \Sigma_{yy} v}}$$

Subsequent pairs: maximize same objective, subject to uncorrelatedness with previous projections

Deep Canonical Correlation Analysis

nonlinear extension of CCA using deep neural networks

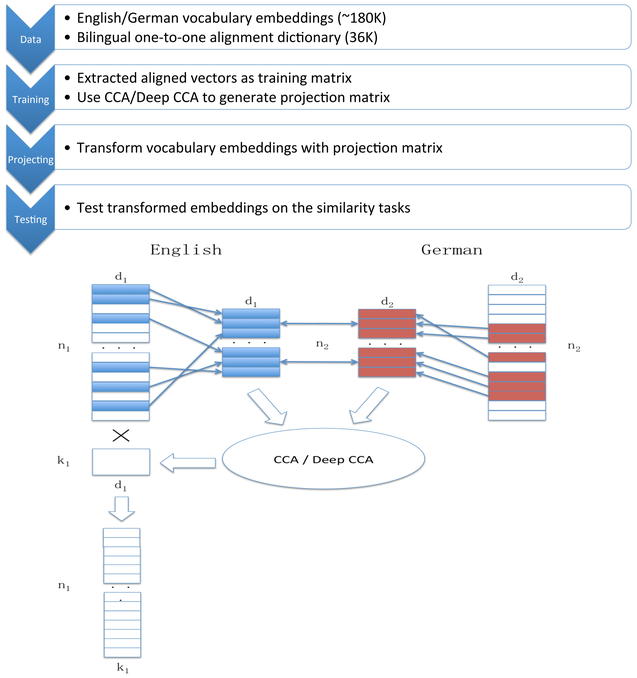
$$\max_{W_f, W_g, u, v} \frac{u^T \Sigma_{fg} v}{\sqrt{u^T \Sigma_{ff} u}\sqrt{v^T \Sigma_{gg} v}}$$



optimization: mini-batch stochastic gradient descent [3]

hyperparameters to tune: output dimension, # layers, layer widths, regularization

Experimental Setup



Results

Two ways of tuning:

BestAvg: maximize average performance on 7 word similarity tasks or 3 bigram similarity tasks
MostBeat: setting that beats baseline on largest number of tasks

Improvement of Deep CCA embeddings on word/ bigram similarity tasks:

Embeddings	WS-353	WS-SIM	WS-REL	SL-999	AN	NN	VN	Avg	Dim
Original	46.7	56.3	36.6	26.5	26.5	38.1	34.1	32.9	640
CCA-1	67.2	73.0	63.4	40.7	42.4	48.1	37.4	42.6	384
CCA-Ens	67.5	73.1	63.7	40.4	42.0	**48.2**	37.8	42.7	384
DCCA-1 (BestAvg)	69.6	73.9	65.6	38.9	35.0	40.9	**41.3**	39.1	128
DCCA-Ens (BestAvg)	**70.8**	**75.2**	**67.3**	41.7	42.4	45.7	40.1	42.7	128
DCCA-1 (MostBeat)	68.6	73.5	65.7	**42.3**	**44.4**	44.7	36.7	41.9	384
DCCA-Ens (MostBeat)	69.9	74.4	66.7	**42.3**	43.7	47.4	38.8	**43.3**	384

Results on bigram tasks when tuned on bigram development sets:

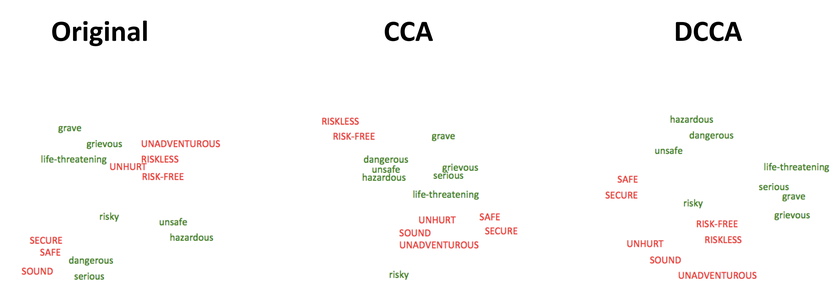
Embeddings	AN	NN	VN	Avg
CCA	42.4	**48.1**	37.4	42.6
Deep CCA	**45.5**	47.1	**45.1**	**45.9**

Discussion

Highly-similar word pairs whose ranks change the most with DCCA:

better with DCCA		worse with DCCA	
arrive	come	author	creator
locate	find	leader	manager
way	manner	buddy	companion
recent	new	crowd	bunch
take	obtain	achieve	succeed
boundary	border	attention	interest
win	accomplish	join	add
contemplate	think	mood	emotion
know	comprehend	god	spirit
population	people	investigate	pursue
student	pupil	crime	violation
begin	go	style	fashion
circumstance	situation	administration	management
sense	intuition	politician	president
give	lend	dumb	dense
acquire	get	realize	discover
competence	ability	pretend	portray
noticeable	obvious	wagon	carriage
cloud	haze	delightful	wonderful
evening	dusk	activity	movement

Visualization
Synonyms and antonyms of *dangerous* using t-SNE



References

[1] Faruqui and Dyer. Improving vector space word representation using multilingual correlation. EACL 2014.
[2] Andrew, Arora, Livescu, and Bilmes. Deep canonical correlation analysis. ICML 2013.
[3] Wang, Arora, Livescu, and Bilmes. Unsupervised learning of acoustic features via deep canonical correlation analysis. ICASSP 2015.